

UDAY G

Security Test Engineer · Mobile & Web Application Security · VAPT Analyst · API Security

udaygopalakrishna@gmail.com · +91 78991-69395 · Bengaluru, India · linkedin.com/in/uday-g-

Professional Summary

Security Test Engineer with 1+ year of experience in mobile application security, VAPT, web application security, and API security testing for banking and financial applications. Experienced in Android APK reverse engineering, static & dynamic analysis, Frida instrumentation, SSL pinning bypass, and OWASP-based security assessments. Skilled in Burp Suite, OWASP ZAP, MobSF, JADX, Apktool, Frida, Nmap, and Wireshark. Reported 50+ vulnerabilities across 20+ production applications and contributed to reducing remediation turnaround time by 30%.

Specialised in Frida-based runtime instrumentation for SSL pinning bypass, root detection bypass, and certificate validation bypass to enable advanced runtime and API traffic analysis in production applications. Proficient in Burp Suite-driven web application and API penetration testing aligned with OWASP Top 10 and OWASP Mobile Top 10 standards. Promoted from Security Test Engineer Intern to full-time within six months based on performance and technical contributions.

Technical Skills

Mobile Application Security: Android APK Security Testing, APK Reverse Engineering, Static Analysis, Dynamic Analysis, SSL Pinning Bypass, Root Detection Bypass, Certificate Validation Bypass, Frida Instrumentation, Runtime Instrumentation, MobSF, JADX, Apktool, ADB, Android Studio

Web Application Security: Web Application Penetration Testing, OWASP Top 10, SQL Injection Testing, Cross-Site Scripting (XSS), CSRF Testing, Broken Access Control Testing, Authentication Testing, Session Management Testing, Input Validation Testing, IDOR Testing

API Security: API Security Testing, API Penetration Testing, REST API Security, Authentication & Authorization Testing, BOLA Testing, Broken Object Level Authorization, API Traffic Interception, Burp Suite, OWASP ZAP

Security Standards & Methodology: OWASP Mobile Top 10 (M1–M10), OWASP Top 10, CVSS Scoring, CVE Analysis, Vulnerability Assessment, Penetration Testing Methodology, Secure SDLC, Threat Modelling, Secure Code Review, SAST, DAST, Security Reporting

Tools & Platforms: Burp Suite, Frida, MobSF, JADX, Apktool, OWASP ZAP, Nmap, Wireshark, Metasploit (Basic), Kali Linux, Ubuntu, Linux CLI, Bash Scripting, Python, Git

Professional Experience

Security Test Engineer

January 2026 – Present

Wizzybox Private Limited · Client: Banking & Financial Domain (Ongoing)

Bengaluru, India · Full-Time

Promoted from intern to full-time Security Test Engineer in six months based on findings quality.

- Performed Android mobile application security testing across 20+ banking APKs covering static analysis, reverse engineering, dynamic testing, API security assessment, and remediation validation.
- Developed custom Frida scripts to bypass SSL pinning, root detection, and certificate validation across 10+ banking APKs.
- Performed APK reverse engineering using JADX and Apktool to identify hardcoded API keys, insecure storage, and business logic flaws.
- Static analysis with MobSF across all applications: exposed Android components, unprotected broadcast receivers, dangerous permission configurations, insecure data storage patterns. **Dynamic analysis with Burp Suite:** identified authentication bypasses, Broken Object Level Authorization (BOLA), IDOR, and injection vulnerabilities in REST APIs.
- Delivered full OWASP Mobile Top 10 (M1 through M10) coverage on every assessment — insecure data storage, insecure communication, improper authentication, insufficient cryptography, code tampering, and extraneous functionality.
- Authored 50+ CVSS-rated vulnerability reports with proof-of-concept steps, business impact analysis, and remediation guidance, contributing to a 30% reduction in MTTR.
- Worked directly with development teams through the remediation cycle — explaining findings, verifying fixes, and formally closing vulnerabilities.

Security Test Engineer — Intern

April 2025 – August 2025

Wizzybox Private Limited

Bengaluru, India · Internship

- Android APK security testing on **5+ mobile applications** using MobSF, JADX, Apktool, and Burp Suite under senior engineer mentorship — building production-level proficiency in static and dynamic analysis from day one.
- Static analysis: identified **insecure data storage, hardcoded credentials, exported activities without access controls, and misconfigured broadcast receivers**. Documented 20+ vulnerabilities with structured findings reports, reproduction steps, and

prioritised remediation recommendations.

- Supported API security testing and dynamic analysis — traffic interception with Burp Suite, correlating runtime behaviour with static findings to build complete, evidence-backed vulnerability reports. **Promoted to full-time within six months** based on report quality and technical output.

Projects

Android Banking APK — Security Assessment Lab — *Frida · JADX · Apktool · MobSF · Burp Suite*

- Performed comprehensive static and dynamic security assessment on Android banking APKs. **JADX for source code analysis, MobSF for automated static scanning, Frida for runtime SSL pinning and root detection bypass.** Found 15+ OWASP Mobile Top 10 vulnerabilities, 10+ hidden API endpoints surfaced after bypass. Hardcoded credentials, insecure SharedPreferences, weak certificate handling all identified and documented.
- API layer testing with Burp Suite after bypass — **authentication weaknesses, broken access control, and insecure data transmission** all found through traffic interception that was not possible before the Frida scripts ran.

Web Application Penetration Testing Lab — *Burp Suite · OWASP ZAP · SQLmap · DVWA · WebGoat*

- Performed web application penetration testing across the OWASP Top 10 using DVWA and WebGoat to identify vulnerabilities including SQL injection, XSS, CSRF, broken authentication, session management flaws, and broken access control.
- Conducted authentication and session security testing including login bypass, session token analysis, privilege escalation, and request manipulation using Burp Suite and OWASP ZAP.

Network Security — Traffic Analysis & Enumeration — *Nmap · Wireshark · Linux CLI · Bash Scripting*

- Network scanning and host enumeration using Nmap — **50+ live hosts and 30+ insecure service configurations** identified across enterprise lab networks. Packet capture and traffic analysis using Wireshark across 1,000+ HTTP, DNS, and FTP packets, flagging unencrypted sensitive data in transit.

Certifications

Learning the OWASP Top 10 — LinkedIn Learning	2026
The OWASP API Security Top 10: An Overview — LinkedIn Learning	2026
Penetration Testing and Ethical Hacking · LinkedIn Learning	2026
Ethical Hacking & Cyber Security with CTF · IIT (ISM) Dhanbad — ChES	2024
Cybersecurity Fundamentals · IBM	2025
Mastercard Cybersecurity Job Simulation · Forage	2025
Deloitte Australia — Cyber Job Simulation · Forage	2025
Tata Cybersecurity Analyst Job Simulation · Forage	2025
Introduction to Cybersecurity · Cisco Networking Academy	2025

Education

Bachelor of Engineering — Computer Science Engineering 2021 – 2025 · CGPA: 7.2 / 10

Maharaja Institute of Technology, Mysuru · Visvesvaraya Technological University (VTU)

Relevant Coursework: Network Security · Cryptography & Network Security · Operating Systems · Data Structures & Algorithms · Database Management · Software Engineering